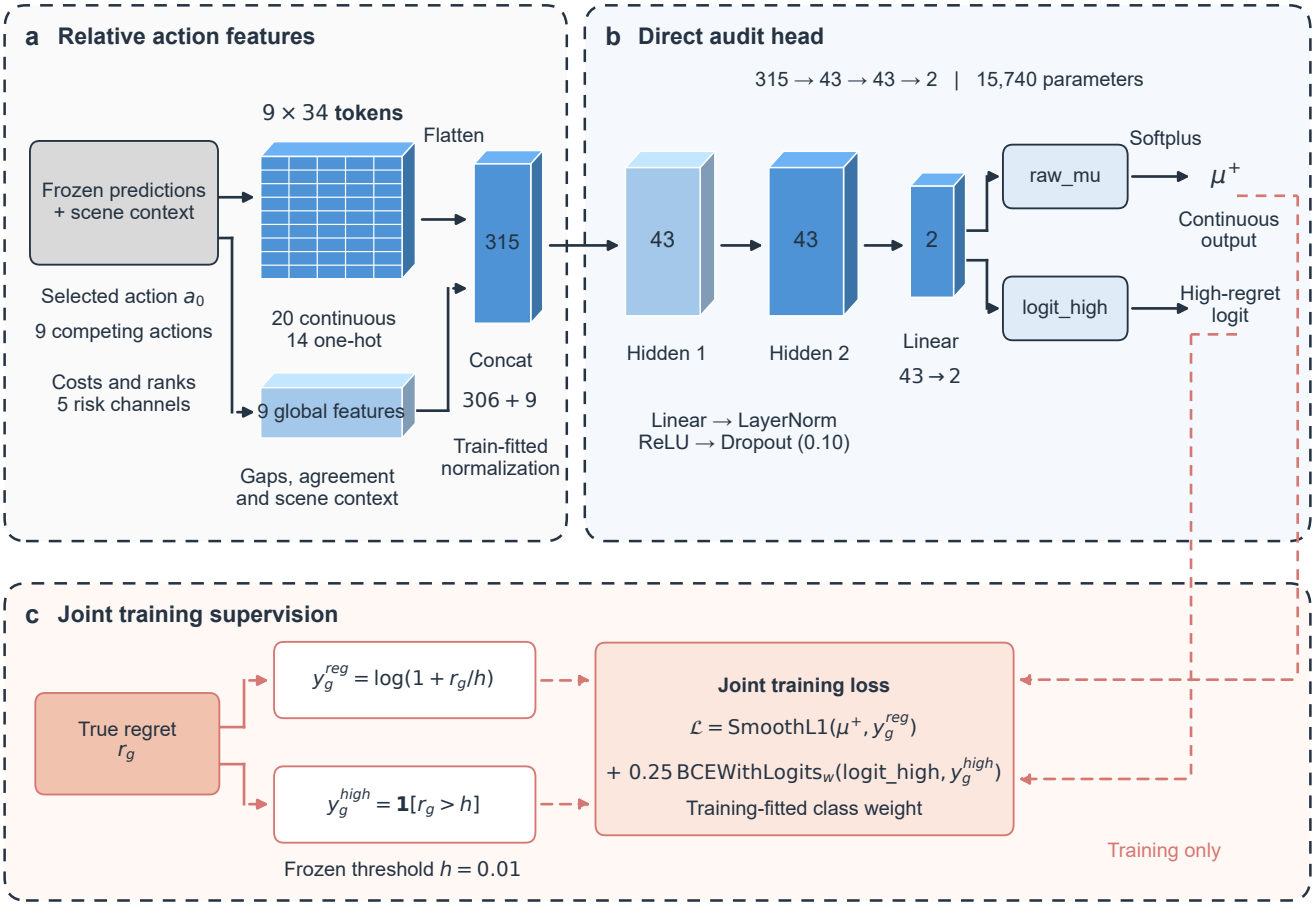

a Relative action features
9 × 34 tokens
Flatten
Frozen predictions + scene context
315
Selected action $a_0$
9 competing actions
20 continuous
14 one-hot
Costs and ranks
5 risk channels
9 global features
Concat
306 + 9
Train-fitted normalization
Gaps, agreement and scene context
b Direct audit head
315 → 43 → 43 → 2 | 15,740 parameters
43
43
2
Softplus
raw_mu
$\mu^+$
Continuous output
logit_high
High-regret logit
Hidden 1
Hidden 2
Linear
43 → 2
Linear → LayerNorm
ReLU → Dropout (0.10)
c Joint training supervision
True regret $r_g$
$y_g^{reg} = \log(1 + r_g/h)$
$y_g^{high} = \mathbf{1}[r_g > h]$
Frozen threshold $h = 0.01$
Joint training loss
$\mathcal{L} = \text{SmoothL1}(\mu^+, y_g^{reg})$
$+ 0.25\,\text{BCEWithLogits}_w(\text{logit_high}, y_g^{high})$
Training-fitted class weight
Training only